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Does the order of the assignments affect the dropout rate?

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***Abstract*— Online learning requires different types of learning support. Learning analytics (LA) studies have focused on support for students. In contrast, few studies use LA to identify learning design elements that could hinder the learning process, i.e., learning bottlenecks. This study aims to find more evidence for this purpose. The quasi-experimental research (n = 228) compares two consecutive course implementations that differ in the assignment order. The assignment order can improve the dropout rate and thus help learning processes. Although the intervention's effect size is small, it is worth performing since it requires low effort.**

Keywords— Learning analytics, Learning bottleneck, Drop out, Learning design

I. Introduction

An increasing amount of higher education teaching and learning is conducted online. The transition from classrooms to online learning sets new challenges for identifying the problems of learning processes. Teachers can observe students' learning processes in classrooms and provide support when needed. Online education makes it more challenging to follow learning processes and to identify the need for timely support.

II. Theoretical Background and Related Work

A. Learning Analytics in Course Design

Learning analytics (LA) has emerged as a critical tool for understanding human learning, teaching, and education by identifying and validating relevant measures of processes, outcomes, and activities [1]. While supporting evidence-based practices derived from assessing learners' progress, research suggests that LA should not promote "one-size-fits-all" approaches, as individual differences strongly influence learning behavior, motivation, and engagement [2].

The field has primarily focused on identifying difficulties in learning processes through various approaches, including predicting final grades [3], identifying at-risk students [4], and evaluating assessment difficulty [5]. Beyond these predictive applications, LA has been instrumental in supporting interventions [6], [7] and enhancing assessment and feedback processes [8].

Integrating LA into educational design processes provides valuable insights for creating personalized and adaptive learning experiences. Recent developments in LA dashboards (LADs) have equipped educators with real-time insights into student performance and engagement patterns [9]. These tools enhance teacher awareness and facilitate informed pedagogical actions, particularly in online learning environments where traditional observation methods are impossible [10].

B. Learning Bottlenecks and Student Progress

Learning bottlenecks represent critical points in a course where students face significant challenges that may impede their progress or lead to dropout. These bottlenecks can manifest in various forms, including cognitive, emotional, and procedural challenges, which disrupt the learning process and can lead to disengagement or failure to achieve learning objectives [11], [12], [13]. Research by [14] revealed that early challenging assignments can create significant bottlenecks affecting overall course completion rates. This finding is supported by [15], who identified course design and assignment structure as key factors affecting student retention in online courses. Current research demonstrates that LA can help identify potential bottlenecks before they impact student success by analyzing the sequences of course assignments.

Sequencing learning activities and assignments plays a crucial role in student success, particularly in online learning environments. [20] conducted a large-scale study across 151 modules, finding that learning design decisions, including assignment sequencing, significantly impacted student behavior and performance. Their research demonstrated that thoughtful activity sequencing could improve student satisfaction and academic performance.

The theoretical foundation for effective assignment sequencing is rooted in scaffolding theory. [21] reviewed a decade of research on scaffolding in teacher-student interactions, highlighting the importance of gradual progression

in task difficulty. This aligns with [18] findings that properly sequenced learning activities improved productive learning behaviors in blended learning environments.

Writing assignments, particularly essays, present unique challenges in online learning environments. [19] emphasizes that the positioning and scaffolding of writing assignments significantly impact student success in online courses. The complexity of writing tasks and their typically high-stakes nature can create pressure points in course design that affect student persistence. Research by [20] demonstrated the importance of proper support structures for writing assignments in online environments, finding that appropriate feedback mechanisms and assignment timing significantly impact student performance. Additionally, [21] identified the transition to academic writing as a particular challenge for students, suggesting that the timing and sequencing of writing assignments could affect success rates.

C. Motivation Theories and Course Sequencing

The relationship between assignment sequencing and student persistence can be understood through Self-Determination Theory (SDT) [22]. SDT posits that intrinsic motivation requires satisfaction of three psychological needs: competence, autonomy, and relatedness. In online learning, the sequencing of assignments directly affects students' sense of competence—their perceived ability to successfully complete tasks.

When challenging assignments appear early in a course sequence, students with lower self-efficacy beliefs may experience what Bandura [23] terms a negative efficacy spiral: initial difficulties diminish confidence, reducing motivation for subsequent tasks. Pajares [24] demonstrated that academic self-efficacy strongly predicts persistence in educational environments.

Research by Hone and El Said [15] found that perceived difficulty of initial tasks significantly predicted subsequent persistence, suggesting that postponing more challenging assignments until students have developed sufficient confidence may improve retention. This is particularly relevant for complex writing tasks, which require higher cognitive load and self-regulation than structured assessments [25]. Hewett and DePew [19] note that writing assignments in online environments create unique motivational challenges due to their complexity and delayed feedback.

D. Research Gap and Study Rationale

While existing research has established LA's value in predicting student success and supporting interventions, there remains a gap in understanding how LA can identify and address learning bottlenecks through course design modifications. While many studies have focused on student-level interventions [26], [27], fewer have examined how LA can inform course design improvements [28].

The literature shows the importance of assignment sequencing [16] and the challenges of writing assignments [19]. However, there is limited research on how LA can inform the optimal positioning of complex writing tasks within online courses. [25] suggest that educators should use LA data to identify elements of learning design that need improvement, but practical approaches for this process are still emerging. This research tries to bridge this gap and thus help educators to optimize learning paths.

This gap is particularly notable given the evidence that course structure and assignment sequence significantly affect student retention [15] and that proper scaffolding can improve learning outcomes [17]. The current study aims to address these gaps by examining how LA can inform the sequencing of course assignments to improve student retention, with particular attention to the positioning of essay assignments within the course structure. The research question for this study is:

RQ1: How does the resequencing of course modules affect student completion rates?

RQ1.1: What is the difference in completion rates between the original and revised sequence?

RQ1.2: How does the position of an essay assignment in the sequence affect its dropout rate?

III. Methods

A. Participants and Context

The study used a quasi-experimental design, following methodological approaches recommended for analyzing natural variations in course design [29], [30]. This design compared two naturalistic course implementations that differed only in the sequence of modules, allowing us to isolate the effect of sequence changes while maintaining ecological validity.

The study was conducted at a Finnish university of applied sciences using data from two consecutive implementations of an online business course focusing on insurance industry expertise. The first implementation (Spring semester) had 115 enrolled students, and the second implementation (Fall semester) had 111 students. The course was delivered through the Moodle learning management system (LMS).

The student data was collected anonymously, including the number of students working on different assignments. The students have given their informed consent to anonymous user data for research purposes. Due to the anonymity of the collected data, no descriptive statistics about the students, such as age, gender, or degree program, are available.

B. Course Structure and Design

The course consisted of 17 assignments organized into three modules, each containing interactive learning content assignments (n=14) and an essay assignment (n=3). Module A covered general topics about the field of business, Module B focused on typical products in the field, and Module C addressed knowledge and skillsets expected from employees in the field. The interactive content was created using H5P and included various assessment types: mini-essays, fill-in-the-blank exercises, and multiple-choice quizzes. The LMS was configured to enforce a linear progression, requiring students to complete each assignment before proceeding to the next one.

For the interactive content assignments, students needed to achieve a threshold of 50% of the total points to complete the assignment. The essay assignments required students to submit their work through the LMS. The weighted mean of the three

essay grades determined the course's final grade, while the interactive learning content assignments were not included in the final grading.

Analysis of the Spring implementation revealed that Module A, which contained an essay about market positioning, resulted in a notably high dropout rate (34.5% of students who started the assignment). Based on this insight, this challenging module's early positioning might contribute to the high dropout rate. Therefore, for the Fall implementation, we rearranged the module sequence to position Module B before Module A while keeping Module C in its original position. This restructuring aimed to test whether allowing students to build confidence through completing potentially less challenging assignments before encountering the identified bottleneck would improve overall course completion.

In both course implementations, students had one semester to complete the course. While the start and end dates were predetermined, students could progress through the assignments at their own pace within this timeframe.

C. Data Collection and Processing

The study utilized anonymized student progression data automatically collected by the Moodle LMS using the Activity Completion report function, following common approaches in LA research [31]. The data was extracted after both course implementations had concluded. For each assignment, we gathered: a) the number of students who started (NS) the assignment (defined as opening the assignment at least once), b) the number of students who submitted the assignment (NSS), and c) the number of students who dropped out during the assignment. The dropout rate (DR) for each assignment was aggregated using the formula:

$$(NS - NSS)/NS = DR$$

D. Data Analysis

We compared the overall completion rates between the two course implementations using a Chi-square test of independence. For several reasons, the Chi-square test of independence was selected as the primary statistical method. First, the data met the key assumptions for this test: the variables were categorical (pass/fail), observations were independent, and expected frequencies were greater than 5 in all cells. Second, we aimed to determine whether the course completion (pass/fail) was independent of the course implementation (first/second), making Chi-square particularly suitable as it tests the independence of categorical variables. The test allows us to determine whether the observed differences in completion rates between implementations could have occurred by chance. Additionally, the Chi-square test is robust to violations of normality, which is important when analyzing educational data that may not follow normal distribution patterns. The analysis included all enrolled students from both implementations (N=226).

The effect size was calculated using Cohen's d to quantify the magnitude of the difference between implementations [32]. Following Cohen's guidelines, effect sizes were interpreted as small (0.2), medium (0.5), or large (0.8).

We identified learning bottlenecks by calculating dropout rates for each assignment, building on previous research in LA that uses dropout patterns to identify challenging course elements [33]. The dropout rate was computed as the ratio of students who dropped out during an assignment to the number of students who started that assignment, a method validated in previous studies of online course completion [34].

To analyze how the repositioning of assignments affected dropout rates, we compared a) the overall progression patterns in both implementations, and b) the transition rates between different types of assignments (interactive content vs. essays). This approach aligns with recent research on learning design that emphasizes the importance of assignment sequencing in online courses [35], [36].

IV. Results

A. Initial Learning Bottleneck Identification

Analysis of the Spring implementation data revealed significant variations in dropout rates across assignments. The most substantial learning bottleneck was identified in Essay 1 (Module A), where 34.5% of students who started the assignment did not proceed to subsequent assignments. Secondary dropout points were observed in Interactive Content 1 (6.3% dropout rate) and Essay 2 (5.9% dropout rate).

Notably, most assignments following the main bottleneck (Essay 1) showed minimal attrition, with several assignments having a null dropout rate. This pattern suggested that once students completed Essay 1, their likelihood of course completion increased substantially.

The analysis of transitions between assignments revealed distinct patterns between implementations. In the Spring implementation, the first essay assignment showed a substantial dropout rate of 34.5%, while interactive content assignments averaged only a 2.8% dropout rate. The Fall implementation demonstrated a more balanced pattern: the first essay's dropout rate decreased to 13.3%, and interactive content maintained a similar average dropout rate of 2.4%. However, the final essay showed a slight increase to 3.1% compared to 0% in the Spring implementation. This represents a notable 21.2 percentage point reduction in the dropout rate for the first essay assignment through resequencing.

B. Effects of Module Resequencing

Based on the identified bottleneck, the Fall implementation reordered the modules to test whether a different sequence would affect student retention. The resequencing moved Module B (focusing on typical products in the field) to the beginning of the course, followed by Module A (general business topics) and Module C (employee knowledge and skillsets).

The module-level analysis (Fig. 1) revealed substantial differences in completion rates between implementations. In the Spring implementation, Module A showed the lowest completion rate at 53.7%, while Modules B and C achieved 94.1% and 100% completion rates, respectively. The Fall implementation, with resequenced modules, showed more balanced completion rates: Module B (now first) achieved 84.7%, Module A (moved to the middle) improved to 93.1%, and Module C maintained a high completion rate of 94.0%. This

represents a notable improvement for Module A, with a 39.4 percentage point increase in completion rate when moved later in the sequence.

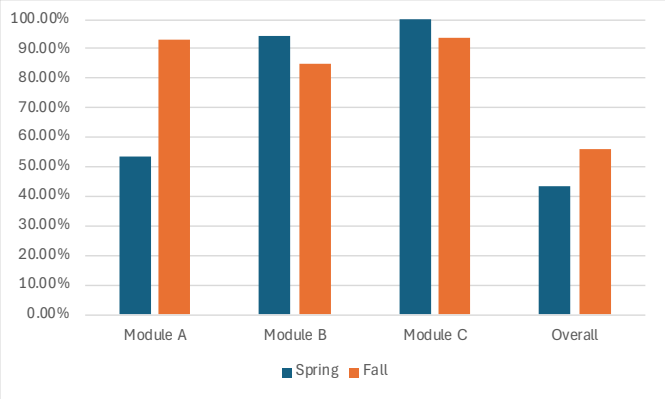


Fig. 1. Comparison of module completion rates between course implementations.

This intervention resulted in markedly different dropout patterns. The dropout rate for Essay 1 decreased dramatically from 34.5% in the Spring implementation to 1.5% in the Fall implementation when positioned later in the course. However, Essay 2's dropout rate increased from 5.9% to 13.3% when moved earlier in the sequence.

C. *Overall Impact on Course Completion*

Statistical analysis of the completion rates between the two implementations revealed significant differences. The Chi-square test of independence yielded $X^2 = 3.94$ ($df = 1$, $p < .05$), exceeding the critical value of 3.841. This result allows us to reject the null hypothesis and conclude that the order of assignments significantly affects course dropout rates.

The effect size, calculated using Cohen's d, was 0.266, indicating the sequence modification's small but meaningful effect. The practical impact is reflected in the completion rates: the Fall implementation achieved a 55.9% completion rate compared to 43.5% in the Spring implementation, representing a 12.4 percentage point improvement.

D. *Module-Level Effects*

The resequencing mainly affected the transition patterns between modules: a) Early Module Progress: In the Spring implementation, 16.5% of students dropped out before reaching the first essay, compared to 15.3% in the Fall implementation, b) Inter-Module Transitions: The Fall implementation showed more consistent progression through modules, with dropout rates more evenly distributed across the course rather than concentrated at specific points, and c) Final Module Completion: In the Spring implementation, all students who began Module C completed it, while the Fall implementation saw a minor attrition (6.2%) in the final module. These findings suggest that while the resequencing successfully mitigated the primary bottleneck, it also created a more gradual pattern of student attrition throughout the course.

V. Discussion

The current research studied if the order of the assignments plays a role in the course dropout rate. As an answer to the research question of to what extent the order of assignments can improve the dropout rate of the business course, the results suggest that the order can have its effect. There is a small effect on the order of the assignments. Still, the effort required from the instructor to change the order is small. This aligns with the previously stated thoughts about executing interventions with small effect sizes if cost-effective [37]. Therefore, the author argues that the order of the assignments should be analyzed with the help of LA. This can help educators to identify the problematic parts of the coursework. The high dropout rate doesn't necessarily mean the assignment is too difficult for students to handle. Reorganizing the assignment into the later part of the course can help reduce the dropout rate of the assignment and the whole course. Changing the order can be sufficient to help students improve their learning process and lead to better dropout rates. The procedure presented in this article is easy to execute for educators with little previous experience with the often-complex methodology used in LA research. Thus, the cost-effectiveness is reasonable.

Despite the content of the assignment, the first Essay assignment seems to be a major obstacle for students. The first assignment requiring more effort from students can be a test measuring motivation [22], [38] or self-efficacy beliefs [24]. When students get past this part of the course, the chances of passing the whole course will increase. In the context of this study, any essay assignment that requires more effort from learners than interactive content assignments can be a learning bottleneck. Yet, the number of students getting past the first essay seems to have a higher opportunity to reach the end of the course and complete each assignment. When the contents and requirements of the assignments remain the same, the order of the assignments should be organized in a way that increases the difficulty or the required effort towards the end of the course.

The findings align with Self-Determination Theory [35], particularly the concept of competence as a fundamental psychological need. The dramatic reduction in dropout rates when the challenging essay assignment was repositioned later in the sequence suggests that students first developed sufficient self-efficacy [36] through completing simpler interactive content before tackling more demanding tasks. This sequencing allowed students to build what Bandura terms "mastery experiences"—successful completions that strengthen confidence for subsequent challenges. The 39.4 percentage point improvement in Module A completion rates when moved later in the sequence demonstrates how early successes foster motivational momentum necessary to persist through complex writing tasks. This supports Keller's ARCS model, which identifies confidence as critical for sustained motivation in learning environments.

The author has assessed the difficulty of the assignments based on the dropout rate. However, other factors could be at play here. For instance, it would be good to know what other courses the students were taking and whether some of the dropouts could be linked to assessment deadlines in other courses. To gain a more comprehensive understanding of the issue, it would be beneficial to collect data from those students who dropped out to ask them why they did. Perhaps it could have been a personal issue or something else. Therefore, it is important to consider other factors that could have contributed

to the dropout rate. Additionally, it would be helpful to consider the performance of the students who completed the assignments to determine the difficulty of the assignments.

The current study is evidence-based research on how LA can help learning design decisions and how students respond to these actions [39]. The practical implications are two-fold. First, the procedure explained in this paper requires the forced order of the assignment. Thus, this should be noted at the initial learning design phase of the online course. Second, once the course is over, the data about the possible bottlenecks accumulates in the later parts or at the end of the course. At the end of the course, the metrics are readily available to detect the bottlenecks and make improvements until the subsequent course implementation. Still, at the early stage of the course, this information is not yet available to help support students. However, it must be noted that this does not remove the need for timely intervention for the students stuck in the bottlenecks during the initial course implementation.

VI. Conclusion and Future Work

This study has experimented with the order of the assignments and its effect on dropout rates in an online course. The results suggest that LA can be used to help analyze this type of question. Furthermore, the results show that the experiment can help improve the dropout rate, and the explained procedure is easy to execute.

While this study provides valuable insights into assignment sequencing, several limitations warrant consideration. The sample size, though sufficient for initial analysis, would benefit from larger-scale replication to strengthen statistical confidence. Additionally, this study focused on a single business course at one institution, limiting generalizability across disciplines with different pedagogical approaches. For instance, STEM courses with scaffolded problem-solving or humanities courses with different writing requirements might show different patterns of bottlenecks. Future research should replicate this approach across diverse disciplinary contexts, institutional settings, and student populations to establish broader applicability of the findings.

The current study puts a unidimensional perspective on the assignment dropout rate. Different metrics, such as the number of times a student opens the assignment before giving up or the throughput time needed to complete it, could shed light on the difficulty of the assignments and help instructors evaluate the need for timely support or rearrange the learning design.

References

[1] K. Mangaroska and M. Giannakos, "Learning analytics for learning design: a systematic literature review of analytics-driven design to enhance learning," IEEE Transactions on Learning Technologies, vol. 12, no. 4, pp. 516–534, Oct. 2019.

[2] D. Gašević, S. Dawson, and G. Siemens, "Let's not forget: Learning analytics are about learning," TechTrends, vol. 59, no. 1, 2015.

[3] A. Namoun and A. Alshanqiti, "Predicting student performance using data mining and learning analytics techniques: a systematic literature review," Applied Sciences, vol. 11, no. 1, p. 237, 2021.

[4] M. Saqr, U. Fors, and M. Tedre, "How learning analytics can early predict under-achieving students in a blended medical education course," Med Teach, vol. 39, no. 7, pp. 757–767, 2017.

[5] M. Nissinen, E. Silvennoinen, and M. Saqr, "How assessment analytics can help to improve reliability, efficiency, and fairness of entrance examinations," in FLAIEC conference proceedings 2022.

[6] M. L. Bernacki, M. M. Chavez, and P. M. Uesbeck, "Predicting achievement and providing support before STEM majors begin to fail," Comput Educ, vol. 158, p. 103999, Dec. 2020.

[7] S. Heikkinen, M. Saqr, J. Malmberg, and M. Tedre, "Supporting self-regulated learning with learning analytics interventions – a systematic literature review," Educ Inf Technol, vol. 28, no. 3, pp. 3059–3088, Mar. 2023.

[8] S. Dawson, A. Pardo, F. S. Kia, and E. Panadero, "An Integrated Model of Feedback and Assessment: From fine grained to holistic programmatic review," in LAK23: 13th International Learning Analytics and Knowledge Conference (LAK 2023), Association for Computing Machinery, Mar. 2023.

[9] T. Ley et al., "Towards a partnership of teachers and intelligent learning technology: A systematic literature review of model-based learning analytics,"Journal of Computer Assisted Learning, vol. 39, no. 5, pp. 1397-1417, 2023.

[10] R. Davies, G. Allen, C. Albrecht, N. Bakir, and N. Ball, "Using educational data mining to identify and analyze student learning strategies in an online flipped classroom," Educ Sci (Basel), vol. 11, no. 11, 2021.

[11] E. Pinnow, "Decoding the Disciplines: An Approach to Scientific Thinking," Psychology Learning and Teaching, vol. 15, no. 1, 2016.

[12] K. B. Kistler, D. E. Tyndall, and A. Peery, "Bottlenecks to Learning Evidence-Based Practice," Nurse Educ, Nov. 2024.

[13] C. J. Stanny, "Overcoming obstacles that stop student learning: The bottleneck model of structural reform.," in Assessing undergraduate learning in psychology: Strategies for measuring and improving student performance., 2020.

[14] M. V. Almeda, J. Zuech, C. Utz, G. Higgins, R. Reynolds, and R. S. Baker, "Comparing the factors that predict completion and grades among for-credit and open/mooc students in online learning," Online Learning Journal, vol. 22, no. 1, 2018.

[15] K. S. Hone and G. R. El Said, "Exploring the factors affecting MOOC retention: A survey study," Comput Educ, vol. 98, 2016.

[16] B. Rienties and L. Toetenel, "The impact of learning design on student behaviour, satisfaction and performance: A cross-institutional comparison across 151 modules," Comput Human Behav, vol. 60, 2016.

[17] J. van de Pol, M. Volman, and J. Beishuizen, "Scaffolding in teacher-student interaction: A decade of research," 2010.

[18] X. Chen, L. Breslow, and J. DeBoer, "Analyzing productive learning behaviors for students using immediate corrective feedback in a blended learning environment," Comput Educ, vol. 117, 2018.

[19] B. L. Hewett and K. E. DePew, Foundational Practices of Online Writing Instruction. The WAC Clearinghouse, 2015.

[20] I. G. Anson, "Assessment Feedback using Screencapture Technology in Political Science," Journal of Political Science Education, vol. 11, no. 4, pp. 375–390, Oct. 2015.

[21] D. A. Jolliffe and A. Harl, "Texts of our institutional lives: Studying the 'reading transition' from high school to college: What are our students reading and why?," College English, vol. 70, no. 6, 2008.

[22] R. M. Ryan and E. L. Deci, "Intrinsic and extrinsic motivation from a self-determination theory perspective: Definitions, theory, practices, and future directions," Contemp Educ Psychol, vol. 61, 2020.

[23] A. Bandura, Social foundations of thought and action : a social cognitive theory, vol. 16, no. 1. Prentice-Hall, Inc., 1986.

[24] F. Pajares, "Self-efficacy beliefs in academic settings," Rev Educ Res, vol. 66, no. 4, 1996.

[25] J. M. Keller, "Development and use of the ARCS model of instructional design," Journal of Instructional Development, vol. 10, no. 3, pp. 2–10, Sep. 1987.

[26] D. Ifenthaler and J. Y.-K. Yau, "Utilising learning analytics to support study success in higher education: a systematic review," Educational Technology Research and Development, vol. 68, no. 4, pp. 1961–1990, Aug. 2020.

[27] J. E. Russell, A. Smith, and R. Larsen, “Elements of Success: Supporting at-risk student resilience through learning analytics,” Comput Educ, vol. 152, no. March, p. 103890, 2020.

[28] S. Maloney et al., “Using LMS Log Data to Explore Student Engagement with Coursework Videos.,” Online Learning, vol. 26, no. 4, pp. 399–423, Dec. 2022.

[29] L. Lockyer, E. Heathcote, and S. Dawson, “Informing Pedagogical Action,” vol. 57, no. 10, pp. 1439–1459, Mar. 2013.

[30] P. Reimann, “Connecting learning analytics with learning research: the role of design-based research,” Learning: Research and Practice, vol. 2, no. 2, 2016.

[31] Z. Papamitsiou and A. A. Economides, “Learning analytics and educational data mining in practice: A systemic literature review of empirical evidence,” Educational Technology and Society, vol. 17, no. 4, 2014.

[32] J. Cohen, “A power primer,” Psychol Bull, vol. 112, pp. 155–159, 1992.

[33] G. Méndez, X. Ochoa, and K. Chiluiza, “Techniques for data-driven curriculum analysis,” ACM International Conference Proceeding Series, pp. 148–157, 2014.

[34] J. L. Santos, J. Klerkx, E. Duval, D. Gago, and L. Rodríguez, “Success, activity and drop-outs in MOOCs An exploratory study on the UNED COMA courses,” ACM International Conference Proceeding Series, pp. 98–102, 2014.

[35] B. Rienties, Q. Nguyen, W. Holmes, and K. Reedy, “A review of ten years of implementation and research in aligning learning design with learning analytics at the Open University UK,” Interaction Design and Architecture(s), no. 33, 2017.

[36] Q. Nguyen, B. Rienties, and L. Toetenel, “Unravelling the dynamics of instructional practice: A longitudinal study on learning design and vle activities,” in ACM International Conference Proceeding Series, 2017.

[37] J. A. C. Hattie, Visible Learning: A Synthesis of Over 800 Meta-Analyses Relating to Achievement. 2009.

[38] T. J. Dunn and M. Kennedy, “Technology Enhanced Learning in higher education; motivations, engagement and academic achievement,” Comput Educ, vol. 137, pp. 104–113, Aug. 2019.

[39] L. P. Macfadyen, L. Lockyer, and B. Rienties, “Learning Design and Learning Analytics: Snapshot 2020,” Journal of Learning Analytics, vol. 7, no. 3, pp. 6–12, Dec. 2020.